FinSight Model Comparison Report

Model Comparison Results (Ranked by Weighted Score: 40% PR-AUC, 30% Recall, 20% ROC-AUC, 10% F1)

1. Logistic Regression - PR-AUC: 0.2146, Recall: 0.6875, ROC-AUC: 0.7473 [Champion Model]
2. LightGBM - PR-AUC: 0.2183, Recall: 0.6158, ROC-AUC: 0.7462 [Challenger Model]
3. CatBoost - PR-AUC: 0.2228, Recall: 0.5038, ROC-AUC: 0.7401 [Discard]
4. XGBoost - PR-AUC: 0.1899, Recall: 0.4257, ROC-AUC: 0.7154 [Discard]
5. Gradient Boosting - PR-AUC: 0.2180, Recall: 0.0141, ROC-AUC: 0.7515 [Discard]
6. Random Forest - PR-AUC: 0.1847, Recall: 0.0008, ROC-AUC: 0.7151 [Discard]
7. Extra Trees - PR-AUC: 0.1840, Recall: 0.0012, ROC-AUC: 0.6985 [Discard]
8. Decision Tree - PR-AUC: 0.0898, Recall: 0.1486, ROC-AUC: 0.5347 [Discard]